

SUMMARY

PhD candidate in privacy-preserving ML with experience in deep learning, LLMs, and full-stack engineering. Track record of delivering research solutions, from hair segmentation models to production RAG pipelines. Seeking a Data Science or ML internship to solve real-world R&D challenges.

TECHNICAL SKILLS

Programming: Python, SQL, C/C++, Java, JavaScript, Bash
ML & Data Sci: PyTorch, Scikit-learn, OpenCV, Pandas, NumPy
Privacy ML: Federated Learning, Differential Privacy
Databases: Vector DBs, MySQL, SQLite, MongoDB
DevOps & Tools: Docker, Git, CI/CD

EXPERIENCE

PhD Researcher — Privacy-Preserving ML
Eötvös Loránd University (ELTE)

Sep. 2025 – Present
Budapest, Hungary

- Design DP-SGD federated learning pipelines for synthetic data ($\epsilon = 1$) and benchmark privacy-utility trade-offs
- Develop the open-source [IDaSec Practicum](#) (8 labs on ML security & privacy) and mentor graduate students
- Received the Stipendium Hungaricum Scholarship for academic excellence and research potential

Software Engineer
RayaneBio

Dec. 2024 – Mar. 2025
Remote

- Architected and deployed a responsive e-commerce platform using React and Next.js for the organic products market
- Implemented secure user authentication and database models, improving data management and transaction security
- Integrated customer feedback loops and testimonial features, elevating brand trust and increasing client engagement metrics

PROJECTS

MyVellum
LLM Orchestration, RAG, Docker, CI/CD

Dec. 2025 – Present

- Architected an LLM orchestration system with modular prompt templates for an AI-powered autobiography platform
- Developed vector-backed RAG pipelines with context-stitching, reducing hallucination rates in generated responses
- Integrated automated evaluation gates into CI/CD for relevance and safety metrics, ensuring quality across deployments
- Owned end-to-end API design and Docker deployment; mentored two engineers on prompt engineering best practices

MS-AGHairNet
Python, PyTorch, OpenCV

Jan. 2025 – Jun. 2025

- Designed a deep learning architecture for hair segmentation, achieving a 95.37% Dice coefficient on the benchmark dataset
- Engineered a data augmentation pipeline that improved generalization across image conditions, reducing overfitting by 15%
- Evaluated model performance with precision, recall, and F1-score; optimized inference pipeline for real-time deployment

EDUCATION

PhD in Informatics — Privacy-Preserving Machine Learning
Eötvös Loránd University (ELTE)

Sep. 2025 – Expected 2029
Budapest, Hungary

Thesis: Synthetic data generation in artificial intelligence security (Supervised by Pr. Lendák Imre)

M.Sc. in Artificial Intelligence (GPA: 3.7/4.0)
University of Khenchela

Sep. 2023 – Jun. 2025
Khenchela, Algeria

Thesis: Hair Dermoscopic Image Segmentation Using Deep Learning (Supervised by Pr. Dalal Bardou)

B.Sc. in Mathematics and Computer Science (GPA: 3.9/4.0)
University of Khenchela

Sep. 2020 – Jun. 2023
Khenchela, Algeria

CERTIFICATIONS & TRAINING

- Software Engineering (Back-end Specialization)** — ALX Africa
- AI Career Essentials (AiCE)** — ALX Africa

Feb. 2025
Jul. 2024

LANGUAGES

Arabic (Native) | English (Professional Proficiency) | French (Intermediate) | Hungarian (Beginner)